

Cannabis in the Capital: Exploring the Spatial Association Between Medical Marijuana Dispensaries and Crime

Abstract

The legalization of medical marijuana remains a controversial policy. An important dimension of marijuana legalization is the siting of dispensaries and their influence on a surrounding community. Specifically, dispensaries, if they engender criminogenic opportunities, have the potential to undermine the social organization of neighborhoods. Using a quasi-experimental design, we examine the relationship between Medicinal Marijuana Dispensaries (MMDs) and crime using micro spatial units in a novel location. Street level geocoded crime data for violent and nonviolent crime patterns before and after the opening of MMD's in Washington, D.C. Crime is analyzed with 100m buffers around each dispensary and control location drawn from propensity score matching. Differences between crime rates pre- and post-operation are compared. With the exception of one location, crime decreased or remained constant in geographical areas following the opening of a dispensary. MMDs do not appear to have an immediate criminogenic effect. Implications for future research regarding potential confounding factors and data limitations are discussed.

Keywords: micro-places; crime; propensity score matching; medical marijuana dispensaries.

Introduction

In 1998, voters passed Initiative 59 in Washington, D.C., legalizing marijuana for medicinal use (Altieri, 2013). The passage of the legislation, however, was stymied for over a decade by the Barr Amendment, which proscribed DC from using appropriated funds DC receives to legalize or reduce penalties for marijuana. In 2011, the Barr amendment was overturned, and in 2013, the first medical marijuana dispensaries were opened. Now, as a spate of states and DC have legalized marijuana in varying capacities, the effects of marijuana legalization foment controversy (Caulkins et al., 2016). Hudak (2014) characterized marijuana legalization as both a social and policy experiment. Some chief questions raised by marijuana legalization, for instance, include the effects legalization will have on property values, public health, traffic deaths, and crime (Bostwick, 2012; Caulkins et al., 2016).

Another salient issue pertaining to the debate of marijuana legalization is the impact of dispensaries themselves. Dispensaries may be hard to site due to concerns about crime and property values (Lofholm & Ingold, 2011; Nemeth & Ross, 2014). Dispensaries, like NIMBY businesses, may be more likely to be sited in economically disadvantaged neighborhoods (Boggess, Perez, Cope, Root, & Stretesky, 2014; Nemeth & Ross, 2014). If medicinal marijuana dispensaries (MMDs) do in fact engender crime, they carry the potential to undermine neighborhood social controls. The limited body of literature examining the association between MMDs and crime is still unclear.

Previous literature studying the spatial relationship between dispensaries and crime have been conducted in a limited number of states (California, Colorado, Oregon, and Washington). The findings from these studies, on the criminogenic effect of dispensaries, varies depending on the units of analysis (e.g., Brinkman & Mok-Lamme, 2017; Contreras, 2017). The objective of

this study is to extend the literature by examining the relationship between MMDs and crime in Washington, D.C. at the micro spatial unit. Using intersections and street segment midpoints as our unit of analysis, we analyzed crime trends at Washington, D.C.'s five dispensaries before and after their opening. Furthermore, we drew several control locations for each dispensary to compare locations with similar demographic and crime characteristics.

Medicinal Marijuana Legislation

The impact of medicinal marijuana legislation (MML) on various forms of crime has been studied throughout the country (Cerdá et al., 2012; Huber III, Newman, & Lafave, 2016; Morris, Teneyck, Barnes, & Kovandzic, 2014; Shepard & Blackley, 2016). These analyses examine changes in crime through the use of panel designs pre- and post-legislation. The consensus is that MML reduces aggregated crime rates (Huber III et al., 2016; Morris et al., 2014; Shepard & Blackley, 2016). However, it is important to note that because crime has been consistently decreasing since the 1990s, there are potential confounding factors that may lead to erroneous conclusions. Also, aggregated data can fail to capture important changes in crime patterns (Wooldredge, 2002). We use the Routine Activities perspective to understand why crime may have shifted from these legislative changes and the implementation of dispensaries.

Routine Activities Theory

According to Routine Activities theory, crime occurs when a motivated offender, suitable target, and the lack of a capable guardian converge in time and space (Cohen & Felson, 1979). Applied to MMDs, this perspective produces equivocal and competing predictions. One conceptualization that may have currency within law enforcement organizations (See California Chiefs of Police Association, 2009) is that dispensaries and their customers serve as suitable targets for motivated offenders. Dispensaries are largely a cash only industry; because legalized

marijuana remains in defiance of federal law under the Controlled Substances Act of 1970, which makes access to bank accounts both difficult and costly (Tighe, 2015). According to Weedmaps (n.d.), an app that assists users in locating dispensaries, three of DC's five MMD's do not accept credit cards. Thus, the abundance of cash and marijuana may make dispensaries, and their customers, a lucrative target. Anecdotal evidence of dispensary robberies suggest that dispensaries are attractive targets (McGhee, 2017; Stimson, 2017). Additionally, customers are suitable targets because they will likely carry cash or marijuana with them and be relatively vulnerable (i.e. geriatric or have debilitating conditions) (O'Connell & Bout-Matar, 2007, Ryan-Ibarra, Induni, & Ewing, 2015). Therefore, MMD customers may be perceived as suitable targets for robbery (Freisthler, Kepple, Sims, & Martin, 2013). Freisthler and colleagues (2017) found that rates for marijuana specific crimes (i.e. crimes where marijuana was the target of the offense or a dispensary was the victim) were positively associated with dispensary density. From these findings and conceptualization, dispensaries may be criminogenic.

Conversely, a position that exponents of marijuana legalization may take is, even if MMD's themselves entice motivated offenders, dispensaries often employ security measures (e.g. security cameras, doormen) that can be effective at preventing crime, at least in areas within a close proximity of a dispensary (Kepple & Freisthler, 2012). While security measures can protect areas around a MMD, dispensaries attract more foot traffic, providing more "eyes upon the street" (Jacobs, 1961; Chang & Jacobson, 2017). Therefore, dispensaries potentially decrease crime because they lead to an influx of capable guardians comprised of the patrons they serve. Washington, D.C.'s zoning and regulations for MMDs is likely to decrease the amount of crime near the dispensary. MMDs are required to use an alarm, outdoor lights, signage, and a safe for storage (Freisthler et al., 2013). Additionally, MMDs are not permitted within residential areas.

D.C.'s five dispensaries are located in commercial areas that may see some benefit from having an increase in "eyes upon the street".

Spatial Analyses

The routine activities perspective offers insightful possibilities that must be examined near the dispensary itself. Taking a more localized approach, several papers have analyzed spatial trends surrounding the construction of medicinal marijuana dispensaries. The majority of spatial analysis on crime related trends surrounding medicinal marijuana dispensaries are located in California (Contreras, 2017; Kepple & Freisthler, 2012; Freisthler, Ponicki, Gaidus, & Gruenwald, 2016), Colorado (Brinkmann & Mok-Lamm, 2017; Freisthler et al., 2017), Oregon, and Washington (Dragone, Prarolo, Vanin, & Zanella, 2017). Of these studies, the overall impact of crime rates surrounding dispensaries and in neighboring census blocks is unclear. It is important to note that the unit of analysis is not consistent among this literature. Varying forms of spatial units have been examined including census block groups (Contreras, 2017; Freisthler et al., 2016; Freisthler et al., 2017), census tracts (Brinkmann & Mok-Lamm, 2017; Kepple & Freisthler, 2012), and counties (Dragone et al., 2017; Hunt, Pacula, & Weinberger, 2018). The relationship between increases in crime and spatial unit is particularly important for consideration. The smaller spatial units reveal an increase in crime (Contreras, 2017; Freisthler et al., 2016; Freisthler et al., 2017), while null findings and negative relationships with crime are supported at the census tract or country level (Brinkmann & Mok-Lamm, 2017; Dragone et al., 2017; Kepple & Freisthler, 2012). Thus, there is clear diversity between MMDs and their relationship to crime. The unit of analysis seems to be a large predictor on the outcome. The relationship between location specificity and crime patterns may reveal some issues with statistical power and findings. Even more concerning is the limited number of empirical studies between medicinal marijuana dispensaries and crime.

While some of the literature is shifting to the impact of recreational dispensaries (Freisthler et al., 2017), the continued study of medicinal marijuana dispensaries and crime may be important for future implementation of MMLs.

The Current Study

The majority of the spatial analyses of MMDs has been conducted in a limited number of areas in the United States. The goal of this paper is to examine the influence dispensaries have on crime using a micro-spatial analysis in Washington, D.C. From the previous literature, we should expect to see an increase in crime at this micro-level, however, this unit of analysis has not been used in marijuana research. Using support from the Routine Activities perspective, we discuss the impact of these dispensaries on the nation's capital.

Data and Methodology

Washington, D.C. is home to a population of around 700,000 individuals. Washington, D.C. has publicly available geocoded crime data from 2008 through the present, available at <http://opendata.dc.gov/>. The last date within the crime data file as of this analysis was 12/31/2017. D.C. provides Part 1 crimes, which approximately include homicides, assault with a deadly weapon, robbery, sex abuse, burglary, motor vehicle theft, larceny [1], and arson. The incident location is only disseminated at either the nearest intersection or the street midpoint, a common way for police departments to disseminate crime information while maintaining some anonymity for the victims (Singleton & Brunsdon, 2014). This aggregation of data prevents us from identifying whether a crime incident is located at the exact address. However, using such micro-level street units are common in current environmental criminology research (Braga, Papachristos, & Hureau, 2010; Weisburd, Bushway, Lum, & Yang, 2004; Wheeler, 2017).

There were five medical marijuana dispensaries open at the time of this research. Table 1 lists those five dispensaries, along with the date they first sold medicinal marijuana. Dates of first sales were obtained from contacting representatives of each store, with the exception of Capital Care. Its first sale date was obtained from Reilly and Wing (2013). Except for Herbal Alternatives, the dispensary dates of sale matched their initial date of operation.

[INSERT TABLE 1 AND FIGURE 1 HERE]

While previous analyses of medical marijuana dispensaries use larger geographic areas, such as census tracts (Brinkman & Mok-Lamme, 2017; Kepple & Freisthler, 2012), census block groups (Freisthler et al., 2016), or census blocks (Contreras, 2017), we conduct analysis at smaller micro places, street midpoints and intersections (what we will refer to as *street units*). Additionally, we conduct analysis in buffers of nearby distances to dispensaries and control locations of 100 to 1,000 meters. There is a total of 21,506 street units within Washington, D.C., including street units with no reported crimes during the study period. See Wheeler (2017) for a further description of how the street units were constructed.

One reason we conduct analysis at micro places is that there are only five dispensaries, and an analysis at larger units is likely to be very underpowered (Oberwittler & Wikstrom, 2009). Examining crime nearby such dispensaries allows for more direct observation of whether crime increases (or decreases) are directly attributable to the opening of the specific dispensary (Weisburd, 2015).

Three of the dispensaries were operational around August 1st, 2013, and two more opened in 2015. When conducting this analysis, it is important to note that such dispensaries may be forced to site in locations that *already* have a higher prevalence of crime (Boggess et al., 2014).

This may be due to current land use zoning, which forces commercial establishments to locate in parcels zoned for commercial establishments. Such locations tend to historically have high crime (Schmid, 1960), and particular micro places generally show consistent crime patterns over long periods of time (Weisburd, et al., 2004). Figure 2 displays a kernel density map of reported crimes from 2008 through 2012 – a time period *before* any of the dispensary locations were opened. One can see that several of the dispensaries were opened in locations with relatively high crime rates towards the central part of Washington, D.C. Two of the dispensaries towards the periphery of the city are in low crime locations. An observation that crime regularly occurs in or nearby several of the dispensaries subsequently does not establish that the dispensaries *caused* crime to increase after they opened.

Additionally, showing crime increased around the dispensary after it opened is not sufficient to establish the dispensary increased crime. For example, thefts have been increasing consistently in D.C. over the sample time frame examined. If one found that thefts increased around a dispensary, it could simply be a result of thefts increasing throughout the city. Subsequently, it is necessary to draw control locations to identify whether crime increased (or decreased) around dispensaries relative to comparable micro places in Washington, D.C.

[INSERT FIGURE 2 HERE]

Drawing Control Locations

With the above consideration in mind, we draw a sample of control locations that have similar crime and demographic characteristics to the dispensary locations *before* they were opened using propensity score matching (Braga, Hureau, and Parapchristos, 2011). [2] First we

predict the probability of being treated among the set of street units. We predict the probability of a street unit i having a dispensary as a function of the count of violent crimes (e.g. assault with a deadly weapon, homicide, robbery, and sex abuse), the count of non-violent crimes (burglary, theft from motor vehicle, larceny, and arson), and a measure of *social disorganization* (Shaw & McKay, 1969). In the equation below the function f is the logistic function, as we use logistic regression to predict the probability of a street unit having a dispensary.

Counts of crimes were restricted to the years of 2008 through 2012, as were counts on street units before the dispensaries opened. The social disorganization measure was constructed via measures in the 5-year American Community Survey estimates for 2013 at the block group level, and then were allocated to street units using proportional areal allocation. This is similar to the procedure validated in Kim (2016), except that the Voronoi tessellation of the street units was used instead of the length of streets (Wheeler, 2017). This is because intersections have no length, so one cannot use the length of street segments as was done in Kim (2016). The variables allocated are then used to construct the social disorganization index are the percentage of female headed households with children, the percentage of families living in poverty, the percentage of renters and vacant households [3], and the percentage of minorities (measured as any individual who is *not* non-Hispanic white). The counts to construct each of these variables were allocated to street units, and then after constructing the percentages a principal components analysis was conducted on the correlation matrix of those demographic variables. The first principle component was extracted and represents the social disorganization index.

Once the equation predicting the probability of a street unit being selected was estimated, we matched control cases that are within 0.01 of the five dispensary locations. Additionally, we restricted matches to within 5 of the total number of violent crimes, within 10 of the total number

of non-violent crimes, and within 0.5 on the social disorganization index. Finally, we eliminated locations within 1,500 meters of a dispensary from being potential control units. This prevents control units from being contaminated by the dispensaries (Braga et al., 2011).

Where possible, we attempted to draw up to five control locations for each dispensary location (Li, Graham, and Majumdar, 2013). Table 2 displays the statistics for the dispensary street units, along with their matched control locations. We are unable to draw a full set of five matches from two of the dispensaries with higher levels of prior non-violent crimes. However, one can see the levels of crime and the social disorganization are reasonably balanced among the dispensary locations and the matched control locations.

[INSERT TABLE 2 HERE]

Results

Given there are only five treated locations, we do not rely on statistical tests (such as t-tests of mean differences) comparing the treated and control locations. Such tests are likely to be very underpowered, even when drawing many control locations. Alternatively, we used statistical graphs to examine any obvious changes in crimes after the dispensaries were opened relative to the control cases in 100-meter buffers around treated and control locations.

First, we calculated the crime rate per year before and after each dispensary opened, using the first operational date for the matched control cases. For example, Capital Care had a total of 235 non-violent crimes in the time period of 1/1/2008 through 7/28/2013 (the day before it opened). This is a total of 2,036 days prior to the sale of medicinal marijuana. We divided 235 by 2,036 and multiplied by 365.25 to get a rate of 42 non-violent crimes per year in the *pre-*

period. Next, we apply the same transformation for the *post* temporal period, which for Capital Care spans from 7/29/2013 through 12/31/2017, a total of 1,617 days. During this time period, Capital Care had a total of 160 non-violent crimes, which produces a rate of 39 per year. Then, finally we subtracted the post rate per year minus the pre-rate per year, to estimate a change in the rate after the dispensary opened. For each of Herbal Alternative's control units, we use the same operation date to determine the time period for the pre and post period.

Next, we conducted this procedure for each of the dispensaries and control locations to plot those differences in a dot plot in Figure 3 (Wilkinson, 1999). In Figure 3, the dispensary locations are plotted as red squares and the control locations are plotted as grey circles. While such a graph cannot tell one whether there is a statistically significant increase or decrease among the changes in the rates of violent or non-violent crimes per year, four of the five locations do not show evidence of crime increases. However, the Metropolitan Wellness Center shows a clear increase in both violent and non-violent crimes relative to its matched control locations. The other locations show evidence of dispensaries potentially decreasing crime (Capital Care for violent and Herbal Alternatives for non-violent) or show consistency with the ranges of the selected control locations.

[INSERT FIGURE 3 HERE]

Such graphs were constructed for crimes at the exact same street unit, as well as larger buffers of 500 and 1,000 meters. The graph for crimes at the exact same street unit were very similar to those using the 100-meter buffer: Metropolitan Wellness Center depicts an increase in crime, whereas the other locations did not. For the larger buffer sizes, none of the changes in

crime rates for the dispensary locations were far outside the changes in crime rates for the control locations (including Metropolitan Wellness Center). There is no clear evidence that dispensaries diffuse crime into the larger surrounding area. Additionally, using a larger unit of analysis, one would not have identified the increase near the Metropolitan Wellness Center.

Examining the specific crime types available, the increase in violent crime for the Metropolitan Wellness Center is strictly limited to robberies, and the increase in non-violent crimes is strictly limited to larceny. No other crime types increased around the Metropolitan Wellness Center after it opened.

Aggregating larcenies within 100 meters around the Metropolitan Wellness Center and plotting the time series of crimes clearly showed an increase in such events temporally approximate to the opening of the dispensary, see Figure 4. There is evidence that crimes were increasing a few months before the dispensary opened. From 2008 through 2012, larcenies did not exceed four for any given month, but in *June* 2013, there was a total of 7 larcenies, and in *July* 2013, there was a total of 6 larcenies. These two months had observably higher numbers prior to the opening of the dispensary.

[INSERT FIGURE 4 AND FIGURE 5 HERE]

A similar time series graph of robberies is not as informative, as robberies only have 0, 1 or 2 in Metropolitan Wellness Center's buffer area per month throughout the time period. There is clear evidence robberies increased; in the pre-time period only 13 of the 67 months (19%) had 1 or 2 robberies, whereas the time period after the dispensary opened 18 of the 53 months had 1

or 2 robberies (34%). However, due to the low count nature, the clarity for when the crime increase occurred is not as apparent for the robberies graph as it was for the larcenies graph.

Figure 5 displays a map zoomed into the location of the Metropolitan Wellness Center, along with a 100-meter buffer. This map shows the dispensary is located within a busy commercial location with many nearby stores and restaurants. Such crime generators are a potential confounding factor for whether the increase in crime is attributable to the dispensary or due to the location having an increased number of individuals congregating nearby (Brantingham & Brantingham, 1995; Eck & Weisburd, 1995, Wilcox, Quisenberry, Cabrera, & Jones, 2004; Wilcox & Eck, 2011). Because crimes are aggregated to street units, we cannot directly attribute any robbery or larceny to the dispensary (and the data also does not distinguish between interpersonal robberies or commercial robberies). Regardless, it is likely that locations tend to diffuse crime into the surrounding area (Bernasco & Block, 2010), making it difficult to determine whether any particular larceny on the street nearby was or was not potentially associated with the dispensary or another nearby store. The area surrounding the Metropolitan Wellness Center has seen several store changes during the timeframe of study. One considerable addition is the opening of an ATM in the buffer area. Outside of this, the other facilities that were changed are primarily restaurant locations. Some of these venue adjustments occur further down the shopping strip, outside of the 100-meter buffer area. These are potentially confounding factors contributing to the changes in crime patterns seen surrounding this dispensary. As previously mentioned, we cannot draw any direct connection between these locales and crimes.

Discussion

This micro place analysis of the medicinal marijuana dispensaries (MMDs) in Washington, D.C. found few crime pattern discrepancies. Of the five locations, the Metropolitan

Wellness Center was the only location to see an increase in crime over the timeframe of study. However, when examining the temporal changes of crime increases around the Metropolitan Wellness Center, larcenies appeared to be increasing slightly before the dispensary opened. Thus, it is possible the increase in robberies and larcenies around that one dispensary may be attributed to changes to the nearby shop fronts. We conclude that the preponderance of evidence in Washington D.C. shows that medicinal marijuana dispensaries have no immediate criminogenic effect on the surrounding area. If the dispensaries themselves were criminogenic, we should see a larger shift in crime directly following the opening of this facility in the area surrounding MMDs relative to similar control locations. The other locations showed a moderate decrease in crime or no change whatsoever.

Our conclusions should be accepted with some reservation. This study is largely limited by the dispensary capacity of Washington, D.C. [at most five dispensaries]. From this limitation, this study is forced to have a small sample size whereas other studies examine a much larger number of opened dispensaries (Contreras, 2017). Dispensaries are also not opened in random locations, but are restricted to particular commercially zoned locations. Thus, we needed to resort to a quasi-experimental analysis and propensity score matching to attempt to draw reasonable control areas. An additional limitation is that the publicly available crime incidents are aggregated to street segment midpoints and intersections in Washington, D.C. This further limits the ability to tie particular crime incidents to a specific location, or the dispensaries themselves.

We found a lack of support for the Routine Activities perspective, particularly for the argument that medicinal marijuana dispensaries would increase crime because they provide vulnerable victims. However, without data on crimes specifically linked to individuals patronizing those MMD establishments, this analysis is unable to make a direct connection

between the facility or patrons and victimization. This limitation for the direct connection between dispensaries and crime is present here and in much of the literature on marijuana dispensaries. As previously mentioned, recent work by Freisthler and colleagues (2017) found an increase in marijuana-specific crime for areas that have a higher dispensary density. We suggest future work should attempt to incorporate marijuana-specific offenses to further understand this relationship, especially in areas similar to Washington, D.C. with a limited number of dispensaries.

Future research should investigate the influence of recreational dispensaries, as there may be potential differences between the two forms of legalized marijuana. For instance, considering there may be differences between recreational marijuana patrons and medical marijuana patients, one group may be perceived as more suitable targets over the other. Ostensibly, the more vulnerable group is medical marijuana patients, as they may be more likely to be geriatric or have debilitating illnesses (Kepple & Freisthler, 2012; Ryan-Ibarra et al., 2015). However, it is possible that recreational dispensaries generate more traffic, and because dispensaries remain a largely cash only industry (see Tighe, 2015), patrons of these stores may be more lucrative targets if not more vulnerable. In sum, then, there are theoretical rationales to believe that medical and recreational dispensaries may have different effects on crime. The novelty of the recreational marijuana industry limits the amount of literature related to this specific relationship (e.g., Dragone et al., 2017; Freisthler et al., 2017). The one article to date that has examined this relationship at the block level (as opposed to the state level), Freisthler et al. (2017), found the density of recreational dispensaries was related to increases in property crime, but not violent crime, in adjacent neighborhoods. Again, however, the geographic units of analysis in this study does not permit the attribution on any reductions in crime to the dispensaries themselves. Future

research should expand upon the work presented here with microspatial analyses and its continued use for both medicinal marijuana dispensaries and recreational facilities. Over half of the country has permitted the use of medicinal marijuana, and its criminogenic impact still remains unclear.

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[1] Larceny (Theft) is further broken down by *theft from motor vehicle* and *theft other* in the DC open data, but it appears such classifications were not consistent with the earlier data series.

Theft from motor vehicle only first appears in 2009 but shows a sharp jump from around 4,000 cases in 2009 to over 8,000 in 2010 (and subsequently rises to around 12,000 cases in 2016).

From 2008 through 2010 other thefts decreased dramatically, from nearly 18,000 in 2008 to around 9,000 in 2010, and then started to rise again to over 14,000 in 2016. These patterns suggest that the cases may not be consistently recorded in the 2009-2010 time period. For this reason, we aggregate them together for the analysis.

[2] An alternative to drawing control locations based on similar crime counts would be to select a particular type of location as a relative control point. For example, Boyd et al. (2012) and Furr- Holden et al. (2016) compare crime around methadone clinics and drug treatment centers to convenience stores. Ratcliffe (2012) compares crime around bars compared to control fire station locations. We do not employ that strategy here, as many such micro places tend to have variable amounts of crime (Eck, Clarke, & Guerette, 2007). So, we select comparable locations in terms of prior crime counts directly.

[3] While social disorganization typically uses the number of individuals who have recently moved in an area as a metric (Shaw and McKay, 1969), this variable is not available in the American Community Survey at the block group level (it is only available at the census tract level). Therefore, we incorporate the percentage of renters in its place, as those renting tend to move more often than those who own their home.